VectorEditGym: Evaluating Specification-Faithful SVG Repair Under Preservation Constraints

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Abstract

Instruction-based vector editing requires two capabilities: making a requested change and leaving everything else alone. The second is easy to miss when an output is judged only as a raster image. We introduce **VectorEditGym**, a compact, difficult benchmark of 40 SVG repair tasks. Each task pairs a corrupted SVG program with a human visual instruction, a hidden target program, 5.05 annotated repairs on average, and an average of 60.55 protected objects. Instructions describe visible defects without exposing element identifiers, coordinates, color codes, or path data. We define a deterministic binary specification reward: requested repairs use attribute-aware visual tolerances, while every unrequested program detail must remain canonically unchanged and the result must be a valid SVG. Canonical target equality is retained only as a diagnostic. Edit completion and Unintended Change Rate (UCR) explain partial outcomes. We evaluate 34 model endpoints (25 listed as open-weight, 5 inexpensive controls, and 4 frontier closed endpoints) over 1360 requests. The strongest endpoint reaches only 10.0% specification success, despite 42.7% requested-edit completion, showing that apparent repair progress and specification-faithful editing remain substantially different. All prompts, outputs, scoring code, costs, and per-task reports are released.

1 Introduction

An instruction such as "the amber signal has gone dark; fix it and leave the station alone" defines both a positive requirement and a large set of negative requirements. The signal must change, while the train, clock, cables, platform, and every unrelated program detail must not. A model can produce a plausible raster while renaming elements, rewriting path data, shifting an unrelated object, or deleting metadata. These are not cosmetic differences for an SVG: they alter an executable, structured artifact (World Wide Web Consortium, 2018).

Most generation metrics emphasize final appearance. That is appropriate for open-ended synthesis, but insufficient for local editing. A repair benchmark must answer three separate questions: Did the requested objects change? Did unmentioned objects remain stable? Is the resulting program valid and usable? VectorEditGym makes all three observable through hidden target SVGs and explicit protected-object sets.

The benchmark is deliberately small and dense rather than broad and shallow. Its 40 tasks contain 202 annotated repairs across station, harbor, campsite, market, and larger scenic illustrations. The public instruction never names a DOM identifier or serialized numeric/path value. Solvers must interpret the visible scene and edit the source accordingly. Figure 1 shows representative inputs and targets.

Our contributions are:

1. a frozen corpus of dense, human-described SVG repairs with explicit intended edits and broad preservation surfaces;

2. a transparent executable evaluator whose binary specification reward combines tolerant requested-edit checks, strict whole-document preservation, and SVG validity;

3. a reproducible 34-endpoint evaluation with response provenance, token usage, cost, failures, and model-produced SVGs; and

4. an empirical account of the gap between completing visible repairs and preserving the full vector program.

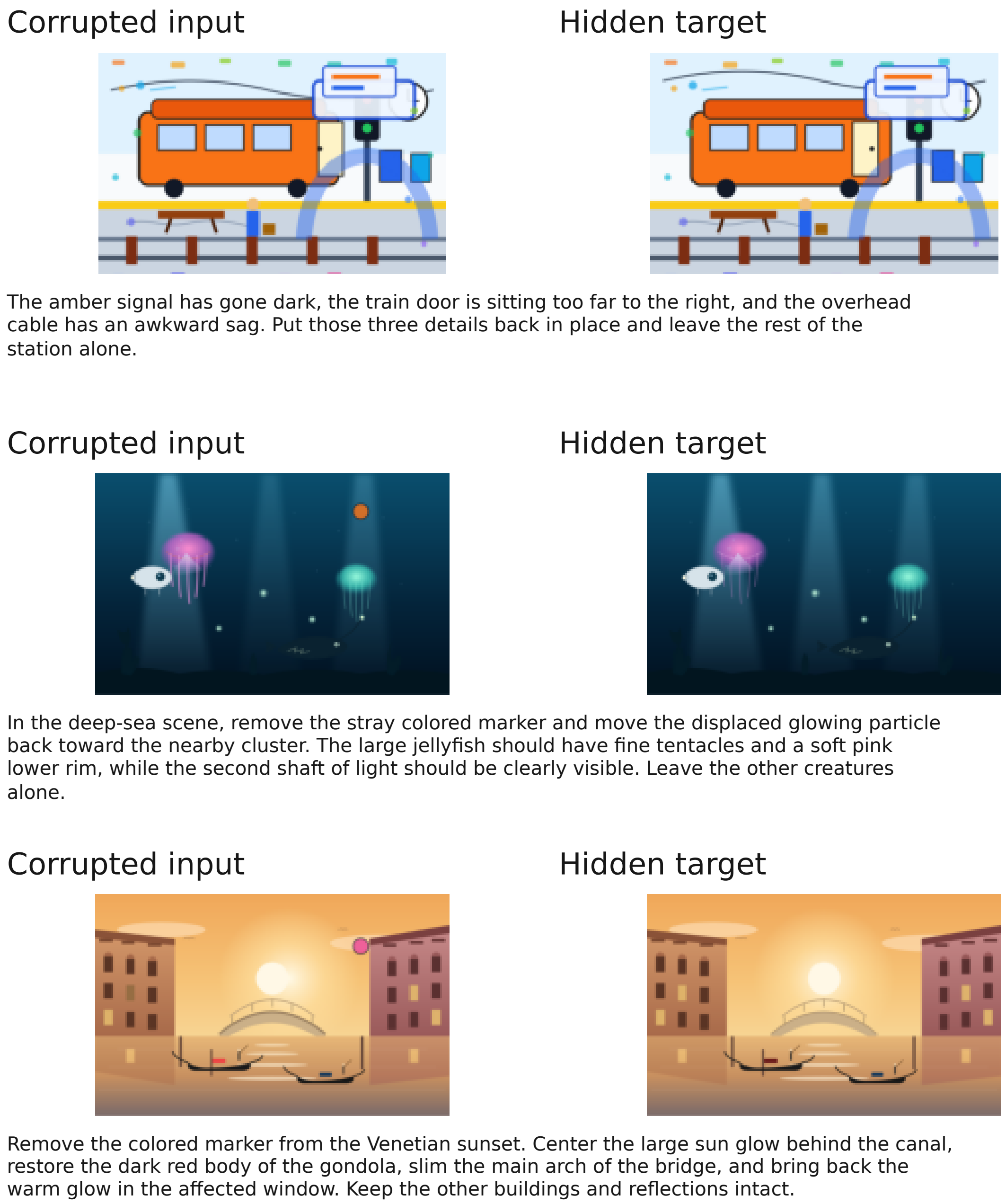


Figure 1: Representative corrupted inputs and hidden targets. Public prompts describe visible mistakes in ordinary language without naming source IDs or revealing target numeric/path values.

2 Related Work

Vector generation. IconShop generates vector icons from text (Wu et al., 2023); StarVector generates SVG code from images and text (Rodriguez et al., 2025); and LLM4SVG studies complex vector understanding and generation with language models (Xing et al., 2025). These systems establish SVG as a useful meeting point between visual generation and program synthesis. VectorEditGym focuses on local repair rather than open-ended generation.

Instruction-based SVG editing. SVGEditBench introduced quantitative evaluation for LLM-based SVG editing (Nishina and Matsui, 2024), and SVGEditBench V2 expanded instruction-based editing evaluation (Nishina and Matsui, 2025). VectorEdits provides a larger dataset and benchmark for vector-graphics editing (Kuchař et al., 2025). Our scope is complementary: we emphasize dense multi-repair scenes, tolerance-aware repair checks, strict preservation outside the requested fields, and publication of every model output. The hidden target anchors deterministic checks without requiring a model to reproduce every requested coordinate or color exactly.

3 Benchmark

3.1 Task Format

Each task is a tuple

$$\tau = (x, S_0, S^*, E, U),$$

where x is a human repair instruction, S_0 is the corrupted SVG, S^* is the hidden target, E is a set of expected object-attribute repairs, and U is the set of object IDs that must be preserved. The model receives only (x, S_0). Target source, expected diffs, target-part lists, and preserve lists are excluded from the prompt and recorded as hidden evaluator inputs.

Instructions were rewritten after task freezing. They describe visible objects and relations—for example, a moon that drifted down and left or a jellyfish whose tentacles became too heavy—without SVG IDs, numeric coordinates, hex colors, path commands, or attribute names. A corpus regression hash covers every non-instruction task field, ensuring this language-only rewrite did not change inputs, targets, annotations, or ordering.

Statistic	Value
Tasks	40
Annotated repairs	202
Repairs per task (mean)	5.05
Protected objects per task (mean)	60.55
SVG elements per task (mean)	85.4
Input characters (total)	294,798

Table 1: Frozen corpus statistics. Element counts include non-rendering SVG nodes; protected-object counts refer to identified scene objects.

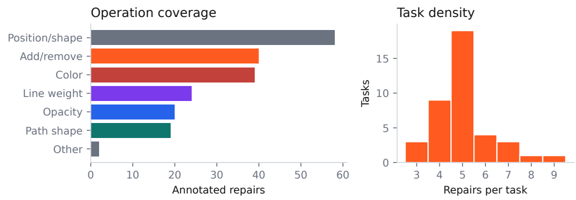


Figure 2: Repair operations and per-task edit density.

3.2 Composition and Difficulty

The corpus contains 20 compact but busy scenes derived from four authored scene templates and 20 larger scenic illustrations. Ten tasks are marked hard and 30 very hard. Tasks require between three and eight repairs, including insertion or deletion, recoloring, displacement, path restoration, line-weight correction, and opacity restoration. All unrequested identified parts are protected.

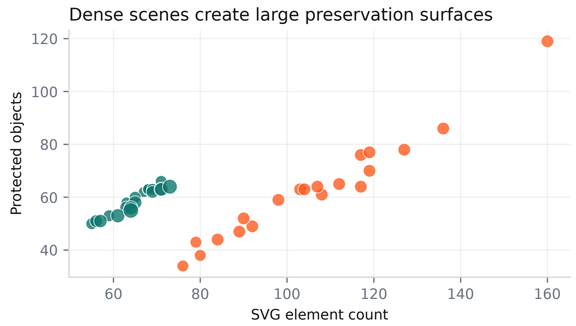


Figure 3: SVG size and preservation surface. Marker area scales with the number of requested repairs.

4 Executable Evaluation

4.1 Three-Gate Specification Reward

Let $C(S)$ parse an SVG and canonicalize representation-only variation. We discard namespace spelling differences and attribute order, normalize numeric spellings and separators in paths, points, transforms, and view boxes, and map common equivalent color spellings (for example, `white`, `#fff`, and `rgb(255,255,255)`) to one form. Element order, nesting, IDs, geometry, paint, text, and other program semantics remain strict. Malformed XML has no canonical form.

For each requested edit $e \in E$, an attribute-aware predicate $\mathrm{ok}_e(S)$ compares the produced value with the hidden target. Existence and non-visual values are exact. Scalar geometry uses absolute error; colors use CIE Lab ΔE_{76}; and point lists and paths use coordinate RMS after requiring the same command topology. When the corrupted-to-target distance is δ_e, the tolerance is bounded by

$$t_e = \min(c_e, \max(f_e, r_e \delta_e), 0.9\delta_e),$$

so the known corrupted value cannot pass. Scalar and color checks use $r_e = 0.35$; path and point checks use $r_e = 0.70$. Position caps are 2% of the relevant viewport dimension, color is capped at $12\ \Delta E_{76}$, opacity at 0.08, and path/point RMS at 2.5% of the smaller viewport dimension. Floors avoid rejecting immaterial serialization noise. These constants are fixed in the released evaluator and surfaced in every per-check report.

Approximation is permitted only at requested fields. Let M_E mask those fields in both the produced and target trees (and remove requested insertion/deletion sites). The strict preservation gate requires $C(M_E(S)) = C(M_E(S^*))$. It therefore catches changes to unlisted definitions, anonymous nodes, root attributes, element order, and unrequested attributes on an edited object, not only drift among named protected objects. The validity gate requires a complete SVG root, well-formed XML, unique nonempty IDs, and syntactically valid normalized attributes.

The primary reward remains binary:

$$\begin{aligned} R_{\text{spec}}(S) = \mathbb{1}&[\text{valid}(S)] \\ &\cdot \mathbb{1}[\forall e \in E : \mathrm{ok}_e(S)] \\ &\cdot \mathbb{1}[C(M_E(S)) = C(M_E(S^*))]. \end{aligned}$$

Canonical target equality $\mathbb{1}[C(S) = C(S^*)]$ is reported separately as a diagnostic and is not required for a pass. No learned judge or model-based grader appears in the public protocol.

4.2 Diagnostic Metrics

For expected checks E, tolerance-aware edit completion is

$$\mathrm{Edit}(S) = \frac{1}{|E|} \sum_{e \in E} \mathbb{1}[\mathrm{ok}_e(S)].$$

For protected objects U, preservation and unintended-change rate are

$$\begin{aligned} \mathrm{Pres}(S) &= \frac{1}{|U|} \sum_{u \in U} \mathbb{1}[C(u_S) = C(u_{S^*})], \\ \mathrm{UCR}(S) &= 1 - \mathrm{Pres}(S). \end{aligned}$$

We additionally report all-repairs pass rate, whole-document clean-preservation rate, SVG validity, normalized source exact match, canonical target match, provider errors, scheduled elapsed time, tokens, and cost. A provider error or invalid output receives reward zero. Diagnostic metrics never replace the binary specification reward.

5 Model Study

5.1 Protocol

We evaluate 25 endpoints designated as open-weight in the study manifest, 5 inexpensive closed controls, and 4 frontier closed endpoints through the OpenRouter chat-completions API (OpenRouter, 2026). These designations record the endpoint grouping used for analysis and are not an independent license audit. The exact endpoint IDs appear in Appendix B.

Every model receives the same system instruction, human repair request, and corrupted SVG source. We record one scored outcome per task, omit temperature rather than imposing a provider-incompatible value, and adapt the output ceiling to input length (minimum 4,096; maximum 12,000 tokens). The runner retries rate limits and transient server failures against the same endpoint. It never falls back to another model; the client permits two transport retries inside each harness attempt. We record requested and resolved model IDs, response ID, finish reason, prompt, completion and reasoning tokens when supplied, scheduled wall-clock elapsed time, and provider-reported cost. Elapsed time begins when a model–task job is scheduled and therefore includes local concurrency-queue wait; we expose it for run auditing, not as endpoint latency. Each source cohort uses a reservation-based budget guard capped at $25; the completed combined study cost $14.88.

5.2 Uncertainty

We compute 95% confidence intervals by resampling the 40 tasks with replacement 10,000 times independently for each model. These intervals quantify task-set uncertainty only. They do not estimate sampling variance because each model–task pair has one scored outcome and no repeated decoding. Endpoint tables are ordered by specification pass, all-repairs pass, edit completion, lower UCR, and model name.

6 Results

Specification success remains rare. Only 20 of 1360 outcomes satisfy all three gates (1.47%), while 679 (49.9%) make at least one annotated repair. Across all outcomes, 1.69% satisfy every requested repair and 19.78% preserve the masked document exactly; only 0.07% canonically match the full hidden target. The strongest endpoint, Claude Sonnet 5, obtains 10.0% specification success. Its edit completion is substantially higher at 42.7%, while its UCR is 38.0% and its SVG validity is 62.5%. The difference is the central benchmark result: models frequently perform some requested visual work but fail the full repair-and-preserve contract. Figure 4 reports task-bootstrap intervals for every endpoint.

Frontier endpoints remain subject to the same bottlenecks. Within the frontier cohort, Claude Sonnet 5 ranks highest with 10.0% specification success and 42.7% edit completion. Its SVG validity is 62.5% and its truncation rate is 40.0%. Length-limited outcomes elsewhere in the cohort remain failures because the protocol requires a complete usable artifact.

Repair and preservation are separate axes. Figure 5 (left) plots requested-edit completion against UCR. A model can move right by correcting visible defects while also moving upward by rewriting protected scene objects. Reporting completion alone would hide this behavior; reporting only full success would hide useful partial capability.

Operation families expose different bottlenecks. Figure 6 decomposes the expected checks. Simple color or add/remove corrections need not predict path-shape or position restoration, especially because the prompt gives visual descriptions instead of literal DOM values. This decomposition also distinguishes an endpoint that attempted an edit but selected the wrong source element from one that returned an invalid or unchanged document.

Cost is not a sufficient capability proxy. The run spans inexpensive small endpoints and larger reasoning or coding models. Figure 5 (right) shows edit completion against total 40-task cost. It is a descriptive endpoint comparison rather than a hardware-efficiency claim: OpenRouter pricing and routing can change, and the study uses one scored outcome per task.

Model	Spec. ↑	Repair ↑	Clean ↑	Edit ↑	UCR ↓	Valid ↑	Cost
Claude Sonnet 5	10.0	10.0	42.5	42.7	38.0	62.5	$2.032
KAT Coder Air V2.5	7.5	10.0	25.0	25.1	57.6	42.5	$0.128
HY 3	7.5	7.5	37.5	52.2	1.8	100.0	$0.092
MiniMax M3	7.5	7.5	25.0	31.2	55.4	45.0	$0.240
DeepSeek V4 Pro	5.0	5.0	20.0	26.8	53.1	47.5	$0.508
Gemma 4 31B IT	2.5	2.5	47.5	49.0	10.8	90.0	$0.084
Devstral 2512	2.5	2.5	57.5	39.7	0.7	97.5	$0.358
Kimi K2.6	2.5	2.5	15.0	15.7	75.1	25.0	$1.077
GPT-5	2.5	2.5	7.5	8.8	85.0	15.0	$1.945
Nemotron 3 Ultra	2.5	2.5	5.0	5.0	92.5	7.5	$0.622

Table 2: Top ten endpoints under binary specification reward. Spec., Repair, Clean, Edit, UCR, and Valid are percentages; cost is the provider-reported cost for all 40 tasks. Full results are in Appendix B.

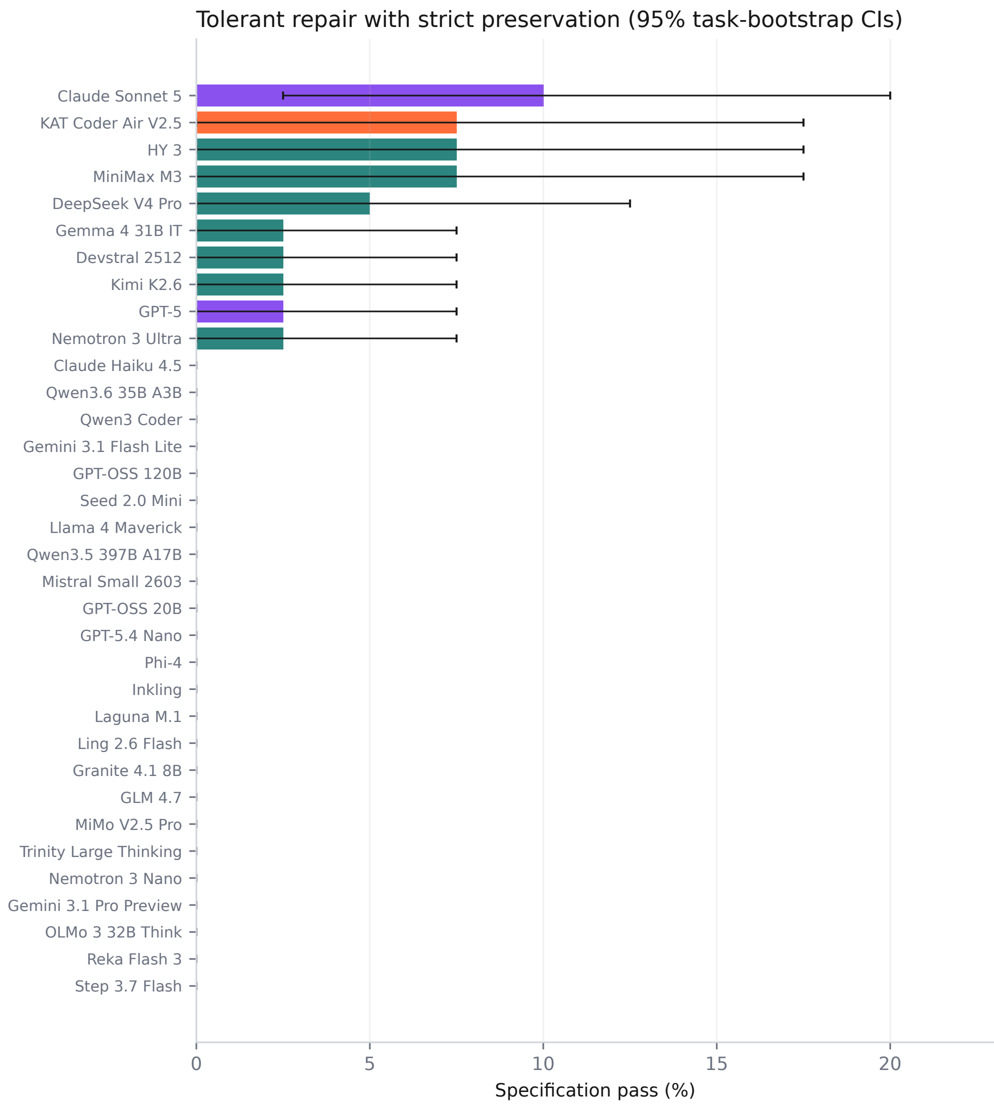
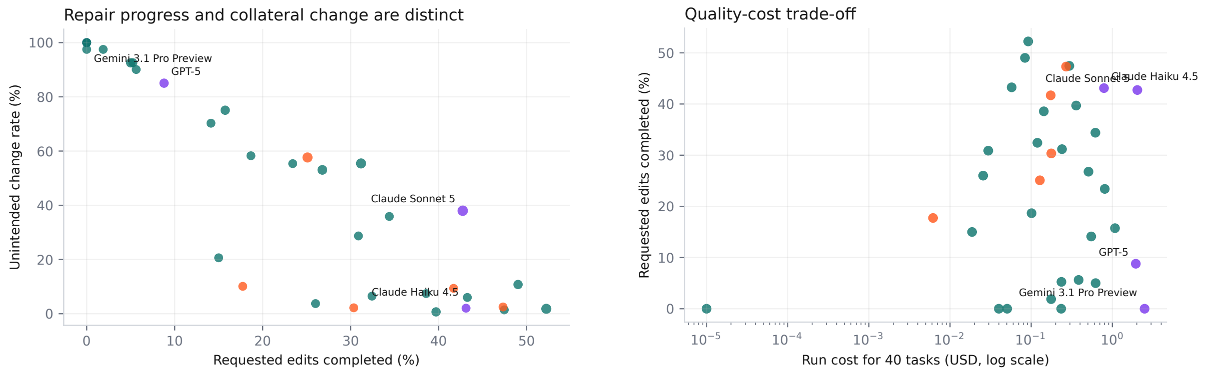


Figure 4: Binary specification success for all endpoints. Orange bars are inexpensive controls, teal bars are endpoints listed as open-weight, and purple bars are frontier closed endpoints.



(a) Repair completion versus unintended change.

(b) Repair completion versus provider-reported cost.

Figure 5: Complementary model-level diagnostics. In (a), marker area scales with specification pass rate and lower right is preferable. Cost in (b) is the recorded total for 40 tasks on a logarithmic axis.

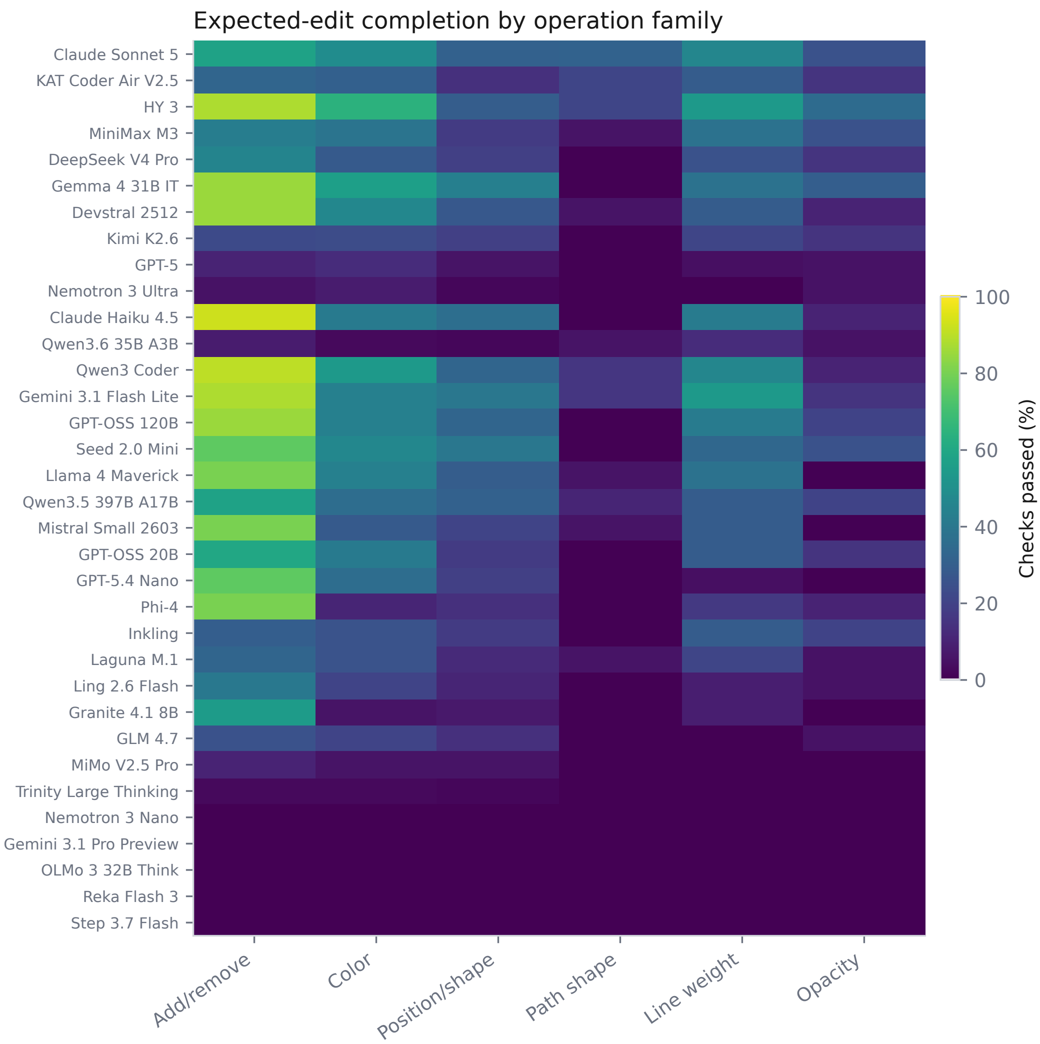


Figure 6: Expected-edit check success by operation family. Rows follow overall specification-pass rank.

7 Error Analysis

Literal partial repair. Models often identify an obvious semantic defect, such as a signal with the wrong color, while leaving a displaced object or malformed curve unresolved. These outputs earn partial edit-completion credit but specification reward zero.

Global rewrites. Some responses regenerate or reformat substantial portions of the SVG. Canonical normalization accepts harmless formatting, but changed paths, reordered scene objects, renamed IDs, or missing definitions count as program changes. Protected object checks identify which named scene components drifted; document-level checks catch anonymous definitions and root attributes.

Approximate repair without collateral change. Prompts intentionally omit exact coordinates, color codes, and path commands. The tolerance-aware repair gate accepts bounded alternatives to the hidden target, but only when the remainder of the SVG is unchanged. This separates a reasonable visual approximation from a global rewrite. The full-target match diagnostic quantifies canonical reproduction without controlling the primary reward.

Provider and truncation failures. Only 54.2% of scored outcomes pass structural SVG validity. The harness preserves raw malformed output, API errors, and finish reasons in the released records. Transient failures are retried only against the requested endpoint; persistent failures remain zero-scored observations. The aggregate provider error rate is 3.0%, and 37.8% of all outcomes end at the provider's output-length limit; parse failure is measured separately.

8 Limitations and Artifact Risks

VectorEditGym has only 40 tasks and should be treated as a focused stress test, not a comprehensive measure of visual editing. Public targets make future contamination possible. The deterministic tolerances are transparent but cannot represent every perceptually acceptable alternative. Path comparison requires matching command topology, CIE76 is only an approximate color metric, and viewport-relative caps may be too strict or permissive for particular scenes. Conversely, masked tree equality does not test downstream animation, scripting, accessibility, or editor-specific behavior.

The evaluation uses one scored outcome per model–task pair and therefore cannot characterize sampling stability. OpenRouter can change prices, routing, model versions, and availability; requested and resolved IDs are published to make this visible. The "open-weight" grouping is manifest metadata, not a legal determination.

Twenty scenic source files in the frozen corpus lack recoverable per-file provenance metadata. We do not claim original authorship of those illustrations. They are retained to preserve the evaluated task set, marked as provenance pending in the artifact, and should be treated as research-only until provenance is resolved. This limits redistribution confidence and is a material artifact risk.

Finally, the benchmark evaluates text-and-source editing rather than multimodal screen interaction. Models see SVG code and a visual description, but not a raster preview or interactive editor. Performance therefore combines scene reasoning, source localization, and code generation under one protocol.

9 Conclusion

VectorEditGym isolates a practical requirement for editing agents: requested change is not enough; preservation is part of the specification. Human visual instructions make the task less like copying a patch, while executable SVG targets make every decision auditable. Across 34 endpoints, partial repair is much more common than specification-faithful repair. The released outputs and diff reports make that gap inspectable at the object and source level and provide a compact target for improving structured visual editors.

A Artifact Notes

Reproducibility. The release includes the frozen corpus, prompts, runner, evaluator, tests, analysis, paper source, Harbor tasks, and per-output viewer. Credentials and transport-level API envelopes are excluded.

Shared authorship. Yug Aditi Gupta and Prannay Hebbar contributed equally to the benchmark, artifact, analysis, and writing. The paper and accompanying material are shared work.

B Full Endpoint Results

Values are percentages. Errors include persistent provider failures and missing usable SVG output. Endpoint IDs, resolved IDs, response IDs, and per-task details are available in the machine-readable release.

Model	Group	Spec.	Repair	Clean	Edit	UCR	Valid	Trunc.	Errors
Claude Sonnet 5	frontier	10.0	10.0	42.5	42.7	38.0	62.5	40.0	0.0
KAT Coder Air V2.5	cheap-control	7.5	10.0	25.0	25.1	57.6	42.5	57.5	0.0
HY 3	open-weight	7.5	7.5	37.5	52.2	1.8	100.0	0.0	0.0
MiniMax M3	open-weight	7.5	7.5	25.0	31.2	55.4	45.0	55.0	0.0
DeepSeek V4 Pro	open-weight	5.0	5.0	20.0	26.8	53.1	47.5	52.5	0.0
Gemma 4 31B IT	open-weight	2.5	2.5	47.5	49.0	10.8	90.0	5.0	0.0
Devstral 2512	open-weight	2.5	2.5	57.5	39.7	0.7	97.5	0.0	0.0
Kimi K2.6	open-weight	2.5	2.5	15.0	15.7	75.1	25.0	72.5	2.5
GPT-5	frontier	2.5	2.5	7.5	8.8	85.0	15.0	85.0	0.0
Nemotron 3 Ultra	open-weight	2.5	2.5	5.0	5.0	92.5	7.5	92.5	0.0
Claude Haiku 4.5	frontier	0.0	2.5	27.5	43.1	2.0	100.0	0.0	0.0
Qwen3.6 35B A3B	open-weight	0.0	2.5	2.5	5.3	92.6	7.5	92.5	0.0
Qwen3 Coder	open-weight	0.0	0.0	45.0	47.5	1.4	100.0	0.0	0.0
Gemini 3.1 Flash Lite	cheap-control	0.0	0.0	25.0	47.3	2.5	100.0	0.0	0.0
GPT-OSS 120B	open-weight	0.0	0.0	42.5	43.3	6.0	95.0	5.0	0.0
Seed 2.0 Mini	cheap-control	0.0	0.0	35.0	41.7	9.4	92.5	2.5	0.0
Llama 4 Maverick	open-weight	0.0	0.0	20.0	38.6	7.4	92.5	0.0	0.0
Qwen3.5 397B A17B	open-weight	0.0	0.0	32.5	34.4	35.9	65.0	32.5	0.0
Mistral Small 2603	open-weight	0.0	0.0	5.0	32.4	6.5	95.0	0.0	0.0
GPT-OSS 20B	open-weight	0.0	0.0	27.5	30.9	28.7	72.5	27.5	0.0
GPT-5.4 Nano	cheap-control	0.0	0.0	22.5	30.3	2.2	97.5	0.0	0.0
Phi-4	open-weight	0.0	0.0	12.5	26.0	3.7	95.0	0.0	0.0
Inkling	open-weight	0.0	0.0	27.5	23.4	55.4	45.0	55.0	0.0
Laguna M.1	open-weight	0.0	0.0	22.5	18.7	58.3	42.5	55.0	0.0
Ling 2.6 Flash	cheap-control	0.0	0.0	15.0	17.7	10.1	90.0	5.0	0.0
Granite 4.1 8B	open-weight	0.0	0.0	10.0	15.0	20.6	77.5	2.5	0.0
GLM 4.7	open-weight	0.0	0.0	12.5	14.1	70.3	30.0	70.0	0.0
MiMo V2.5 Pro	open-weight	0.0	0.0	5.0	5.6	90.1	10.0	87.5	0.0
Trinity Large Thinking	open-weight	0.0	0.0	0.0	1.9	97.5	2.5	97.5	0.0
Nemotron 3 Nano	open-weight	0.0	0.0	0.0	0.0	97.5	0.0	80.0	0.0
Gemini 3.1 Pro Preview	frontier	0.0	0.0	0.0	0.0	100.0	0.0	100.0	0.0
OLMo 3 32B Think	open-weight	0.0	0.0	0.0	0.0	100.0	0.0	0.0	100.0
Reka Flash 3	open-weight	0.0	0.0	0.0	0.0	100.0	0.0	12.5	0.0
Step 3.7 Flash	open-weight	0.0	0.0	0.0	0.0	100.0	0.0	100.0	0.0

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